CECS 229 – Assignment 3

Prime Functions

Given the file, 'pa3.py', complete the code for the following functions:

1. `primes(a, b)` – this function returns a list of all prime numbers between a and b, inclusively, by using the Sieve of Eratosthenes.
    a. a – is the lower bound, which is a positive integer that is greater than 1. The function should throw a ValueError exception if a is less than 2.
    b. b – is the upper bound, which is a positive integer that is greater than or equal to a. The function should throw a ValueError exception if b is less than a.
    c. The returned value should be an integer list of all primes that are between a and b, inclusively.
    d. Iterate through possible divisors and iterate through the range of possible primes using either nested for loops or list comprehensions and test to see if the value should be removed from the list.
    e. *Note* depending on how you implement this function, you may want to make a copy of the list you're removing from, because iterating over a list that is being modified can have unexpected results.
2. `gcd(a, b)` – this function calculates and returns the greatest common divisor of a and b using the Euclidian algorithm.
    a. a – is a non-zero integer (positive or negative, but not zero). The function should throw a ValueError exception if it is 0.
    b. b – is a non-zero integer that is less than or equal to b. The function should throw a ValueError exception if it is 0 or if b > a.
    c. The returned value should be an integer of the gcd of a and b.
    d. Use a loop to repeatedly reduce the gcd by finding the remainder of a and b until the remainder is 0.
3. `bezout_coeffs(a, b)` – this function calculates and returns the bezout coefficients s and t of a and b using a modification of the Extended Euclidian Algorithm.
    a. a – is a positive integer > 0.
    b. b – is a positive integer > 0.
    c. The returned value should be an integer tuple (s, t).

Example Input/Output:

```
primes(37, 55)
[37, 41, 43, 47, 53]
primes(24, 47)
[29, 31, 37, 41, 43, 47]
gcd(53, -19)
1
gcd(54, 48)
6
bezout_coeffs(6, 81)
(-13, 1)
bezout_coeffs(30, 84)
(3, -1)
```

Notes:

1. Please place your name in the comment block at the top of the provided file.
2. Fill in the necessary code to complete the functions. Write your code where specified (ie. the TODOs). Do not modify the file name, function names, or parameters otherwise the program and tests will not work. Return the results in the format specified (ie. correct types and container types), otherwise the program and tests will not work properly.
3. Your solutions may not use any built-in functions that perform the equivalent task (ex. math.gcd()).
4. Test your code using the provided 'main.py' file. You can write additional tests in the 'your_tester.py' file to further check your solutions. Note: just because the main tests pass does not guarantee that it will pass the tests on Gradescope.
5. When you are finished, submit your 'pa3.py' file to Gradescope on Canvas for further testing and grading. It should allow you to make repeated submissions.